

TASKORA

Team Management Application

Sprint 3 Retrospective Report

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1. Sprint Overview

Sprint 3 ran from 12 May to 19 May 2026, covering eight days. The sprint focused on extending Taskora with real-time capabilities and collaboration features, building on the task management and workspace foundations established in Sprints 1 and 2.

Sprint 3 delivered a complete set of real-time and collaboration features: the Socket.io server with all 7 notification triggers, overdue detection cron, deadline reminders, comment CRUD endpoints, calendar and contribution endpoints on the backend; and the notification bell, threaded comment UI, calendar page, contribution page, and activity log page on the frontend. The sprint also closed all five action items carried forward from Sprint 2, including the invite system redesign, edit task UI, status lock enforcement, token expiry fix, and email field clear button. Email verification on account registration was also completed this sprint.

2. Sprint Deliverables

2.1 Completed Tasks

Task ID	Description	Story Points	Status
S3-01	Setup Socket.io server — JWT verify on connect, join workspace-id & user-id rooms	1	Done
S3-02	Implement all 7 notification triggers and push via Socket.io	0.5	Done
S3-03	Implement notifications MVC — notification/:id/read, /read-all, /user/notifications	0.5	Done
S3-04	Implement overdue cron (hourly) + fallback check on every GET /tasks	0.5	Done
S3-05	Implement deadline reminder notifications — push via Socket.io at 24h & 1h before deadline	0.5	Done
S3-06	Implement POST /tasks/:taskId/comments — comment & reply, reject reply-to-reply → 400	0.5	Done
S3-07	Implement GET /comments: BATCH & DELETE comment (author/leader)	0.5	Done
S3-08	Implement GET /workspaces/:id/calendar — tasks with start_date & deadline	0.5	Done
S3-09	Implement GROUP contributions — GROUP BY: task count & % share total: 100%	0.5	Done
S3-10	Implement final contribution summary on archive (paginated) — GET activity	0.5	Done
S3-11	Add activity_log triggers: shortcut_added, comment_added	0.5	Done
S3-12	Connect Socket.io to Navbar / Sender — join rooms, auto-reconnect	0.5	Done
S3-13	Build real-time bell icon, paginated notification list	1	Done
S3-14	Build threaded comments per task — post, reply (1 level), auto-scroll	1	Done
S3-15	Build workspace calendar page — tasks by start_date & deadline	1	Done

Task ID	Description	Story Points	Status
S3-16	Build contribution data — task count & percentage per member	0.5	Done
S3-17	Build activity log page (paginated), integrate all Sprint 3 APIs	0.5	Done
S3-18	Review Sprint 3 deliverables against AC: real-time, sort calendar, contributions, comments	—	Done
S3-19	Conduct Sprint 3 stakeholder demo; document feedback; refine backlog for Sprint 4	—	Done
S3-20	Integration tests: Socket.io, all 7 notification triggers, overdue cron, deadline reminders	—	Done
S3-21	Test comment CRUD: reply-to-reply rejection, document defects, regression tests Sprint 1–2	—	Done
S3-22	Test activity log triggers and pagination, document defects, regression tests Sprint 1–2	—	Done
S3-23	Produce Sprint 3 retrospective report — what went well, what didn't, action items for Sprint 4	—	Done
S3-24	Fix DF-05: Add edit task button and pre-filled form on the task detail page, calling the existing PATCH endpoint	1	Done
S3-25	Fix DF-01: Display clear error message on frontend when invite-by-email returns 404; backend validation already rejects unregistered emails	0.5	Done
S3-26	Fix DF-04: Update the token handling to reflect the corrected 7-day refresh token expiry	0.5	Done
S3-27	Redesign invite system: replace shareable link with a short invite token/code that the leader shares manually; invitee inputs the code on the join workspace page	1	Done
S3-28	Fix DF-02: Add a clear/remove button to the email field in the create workspace form	0.5	Done
S3-29	Enforce one-way status transitions: prevent tasks from reverting from Done once verified, as flagged by Stakeholder 1	0.5	Done
S3-30	Add a 'today' indicator to the calendar view (dot or colour highlight), as requested by Stakeholders 2 and 4	0.5	Done
S3-31	Differentiate overdue badge and deadline label colors, e.g. red for overdue, blue for deadline, as flagged by Stakeholder 2	0.5	Done
S3-32	Implement email verification on account registration	—	Done
S3-33	Improve overall UI polish: make the design more visually compelling, e.g. better colour use, component styling, and visual hierarchy	1	Done

Total story points completed: 16.5 | 32 tasks: Done

2.2 Acceptance Criteria Review

Sprint 3 deliverables were reviewed against the Acceptance Criteria defined in the Product Backlog (Notion) as part of task S3-18. All tasks passed their respective acceptance criteria. No outstanding failures were carried forward into Sprint 4.

3. Stakeholder Demo

3.1 Demo Format

A live demo session was conducted with five stakeholders, all university students representing the target users of Taskora. Stakeholders responded to twelve structured feedback questions after the session.

3.2 Feedback Summary

Overall satisfaction averaged 4.5 out of 5. All five stakeholders rated the sprint at 4 out of 5 or higher.

The redesigned invite code system was positively received by three of the five stakeholders, though two preferred the previous link-based approach for requiring no manual typing. Two stakeholders also noted that the invite code felt too long and suggested either shortening the code or providing a shareable link alternative. The edit task flow was confirmed as smooth and easy to discover by all five stakeholders. Status lock on verified Done tasks was confirmed working by all stakeholders, directly resolving the most critical issue from Sprint 2. The shared calendar with the new today indicator was found clear and useful across the board. The contribution page layout was considered fair and readable at a glance, with a minor suggestion to round decimal percentages for cleaner presentation. The threaded comment section UI was unanimously confirmed as intuitive and familiar. Overdue versus deadline colour differentiation was confirmed as clearly visible by all stakeholders, successfully addressing feedback raised in Sprint 2.

The primary functional concern raised during the demo was the notification panel, which failed to function for two stakeholders due to incomplete backend Socket.io trigger integration. Two stakeholders independently suggested adding colour-coded indicators to the activity log based on action type to improve readability. Stakeholder 2 also noted that switching workspaces from the activity log required unnecessary navigation and suggested an inline workspace selector for smoother usage. Stakeholder 3 requested an unverify capability for leaders on verified tasks. Additional usability feedback included improving the visibility of small UI text elements, particularly the sign-in button text and project labels under the activity log. Stakeholder 3 also suggested enhancing the assigned task page to support viewing assigned tasks across all workspaces with optional workspace filtering. Stakeholder 4 confirmed full satisfaction with the sprint, while Stakeholder 5 noted that the overdue colour improvements requested in Sprint 2 had been successfully implemented.

4. Retrospective

4.1 Highlights

- All planned Sprint 3 core tasks were completed within the sprint window. Email verification was not completed and will be carried forward to Sprint 4.
- All five Sprint 2 action items within scope for Sprint 3 were fully resolved: DF-04 (token expiry), DF-05 (edit task UI), DF-06 (invite redesign), DF-07 (email field clear button), and DF-08 (status lock enforcement).
- The full backend real-time stack was delivered, including the Socket.io server, all seven notification triggers, overdue cron processing, 24-hour and 1-hour deadline reminders, threaded comment CRUD operations, shared calendar endpoint, and contribution calculation system.
- Average stakeholder satisfaction increased from 3.8 / 5 in Sprint 2 to 4.5 / 5 in Sprint 3, representing the highest score achieved across all sprints to date.

- Status lock enforcement on verified Done tasks was confirmed working by all five stakeholders during the live demo session.
- Overdue versus deadline colour differentiation requested in Sprint 2 was successfully implemented and positively acknowledged by stakeholders.
- The threaded comment section UI received unanimous positive feedback, with stakeholders describing the interface as intuitive and familiar.
- The shared workspace calendar, including the today indicator feature, was well received by all stakeholders and considered useful for deadline visibility.

4.2 Challenges & Issues

- Notification Panel Instability During Demo

Two of five stakeholders reported the notification panel did not render during the live demo session. Investigation points to a z-index layering issue (the panel rendering behind other content). The panel was confirmed working for three stakeholders, suggesting the issue is environment-dependent or session-specific. This is documented as Defect 1 in the Sprint 3 defect report and must be stabilised before Sprint 4 UAT.

- Invite Code Design Trade-off

The invite system was redesigned from a shareable JWT link to a short code, addressing DF-03 (irrevocable invite links). However, two stakeholders, Stakeholder 3 and Stakeholder 4, indicated they preferred the original link format because it required no manual typing of a code and website URL. Stakeholder 2 additionally noted the code still feels longer than expected. This represents a partial regression in perceived usability for a subset of users and will be reassessed in Sprint 4, either by shortening the code further or offering a link option alongside the code.

- Contribution Percentage Display

Stakeholder 2 noted that contribution percentages displayed as unrounded decimals feel imprecise and odd. This is a minor polish item but affects perceived quality of the contribution feature. Rounding to whole numbers or one decimal place is targeted for Sprint 4.

- Activity Log Usability Feedback

Two stakeholders independently suggested adding colour-coded indicators to the activity log based on action type to improve readability and activity recognition. Additional feedback noted that switching workspaces from the activity log currently requires unnecessary navigation. An inline workspace selector or dropdown filter was suggested to reduce navigation friction.

- Additional UI and Workflow Improvements

Stakeholders also provided several smaller usability recommendations, including improving the visibility of small UI text elements such as the sign-in button text and project labels under the activity log. Stakeholder 3 additionally requested an unverify capability for leaders on verified tasks and suggested enhancing the assigned task page to support cross-workspace task visibility with optional workspace filtering.

4.3 Lessons Learned

- Stakeholder feedback on UI redesigns can reveal usability regressions not captured by acceptance criteria. Validating redesigns informally with at least one stakeholder before sprint close would reduce late-stage surprises.
- Real-time features (Socket.io, notification triggers) require careful end-to-end testing before demos to avoid environment-specific rendering failures. A dedicated pre-demo smoke test checklist would prevent issues like the z-index defect from surfacing live.

5. Integration Testing

5.1 Scope

Integration tests were executed for completed Sprint 3 tasks as part of tasks S3-20, S3-21, and S3-22. Testing covered the Socket.io frontend connection, notification panel UI behaviour, comment section UI (including reply-to-reply rejection), calendar page rendering, contribution page layout, activity log pagination, and regression tests for Sprint 1 and Sprint 2 features. The test case details are to be filled in below.

5.3 Defect Log

The following defects were identified during Sprint 3 integration testing. Given proximity to the sprint close date, all fixes have been deferred to Sprint 4.

ID	Description	Severity	Target Sprint
DF-01	Notification panel renders behind other content; caused by incorrect z-index or overflow: hidden clipping context on parent element	Medium	Sprint 4
DF-02	Verification status inconsistent between task detail page and assigned task view; views maintain separate local states instead of a shared source of truth	High	Sprint 4
DF-03	End date can be set before start date; no client-side or server-side validation prevents invalid date ranges, corrupting task timeline data	Critical	Sprint 4
DF-04	Task can be created without a start or due date; both fields are not enforced as required, undermining scheduling, deadline tracking, and overdue detection	Critical	Sprint 4
DF-05	Members cannot leave a workspace after joining; no Leave Workspace option exists, creating a usability gap and a privacy concern for users who joined by mistake	High	Sprint 4
DF-06	No option to unverify a verified task; accidental or incorrect leader verification permanently locks the task with no recovery path	High	Sprint 4

6. Action Items for Sprint 4

	Action Item	Target Sprint
1	Fix DF-01: Apply a high z-index to the notification panel and ensure it uses position fixed or absolute outside any clipping parent; test across all pages for stacking context conflicts	Sprint 4
2	Fix DF-02: Ensure task detail page and assigned task view share a single source of truth (same API call or shared state store); trigger a refresh in all relevant views on any verification change	Sprint 4
3	Fix DF-03: Add client-side date picker constraint (minimum due date = selected start date) and server-side validation that rejects submissions where due date precedes start date, with a clear error message	Sprint 4
4	Fix DF-04: Enforce start date and due date as required fields with visible indicators in the form; prevent submission if either field is empty	Sprint 4
5	Fix DF-05: Add a 'Leave Workspace' option in workspace settings with a confirmation dialog; handle edge case where the leaving member is the sole leader by prompting ownership transfer	Sprint 4
6	Fix DF-06: Add an Unverify button (leaders only) on verified tasks	Sprint 4
7	Reassess invite system: shorten the invite code and provide a shareable link alongside the code to address usability concerns	Sprint 4
8	Round contribution percentages to whole numbers for cleaner display, as requested by Stakeholder 2	Sprint 4
9	Add colour-coded dot indicators to the activity log by action type	Sprint 4
10	Improve activity log navigation by adding an inline workspace/project dropdown filter to reduce switching friction	Sprint 4
11	Increase sign-in button text size and improve project label visibility under the activity log for better readability	Sprint 4
12	Enhance assigned task page to support cross-workspace task aggregation with optional workspace filtering	Sprint 4
13	Implement email verification on account registration	Sprint 4